PAPER SUMMARY

AI-Driven Merchant Trust Infrastructure for Rapid Onboarding in Global Digital Marketplaces

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ABSTRACT

Digital marketplaces depend on rapid merchant onboarding while maintaining strong fraud prevention. Traditional onboarding systems rely on static verification techniques such as document validation and rule-based checks, which are increasingly ineffective against sophisticated fraud strategies including synthetic identities, credential compromise, and coordinated fraud networks.

This article presents an AI-driven merchant trust infrastructure that integrates identity verification, behavioral analytics, device intelligence, and graph-based fraud detection into a continuous trust evaluation framework. The proposed system replaces static onboarding checkpoints with probabilistic trust scoring that evolves over time.

The architecture enables scalable marketplace growth by reducing onboarding friction for legitimate merchants while improving detection of high-risk entities. Experimental evaluation demonstrates improvements in fraud detection accuracy, reduced false positives, and increased approval rates.

This work establishes continuous trust systems as a foundational capability for secure and scalable digital marketplaces.

HIGH-LEVEL ARCHITECTURE

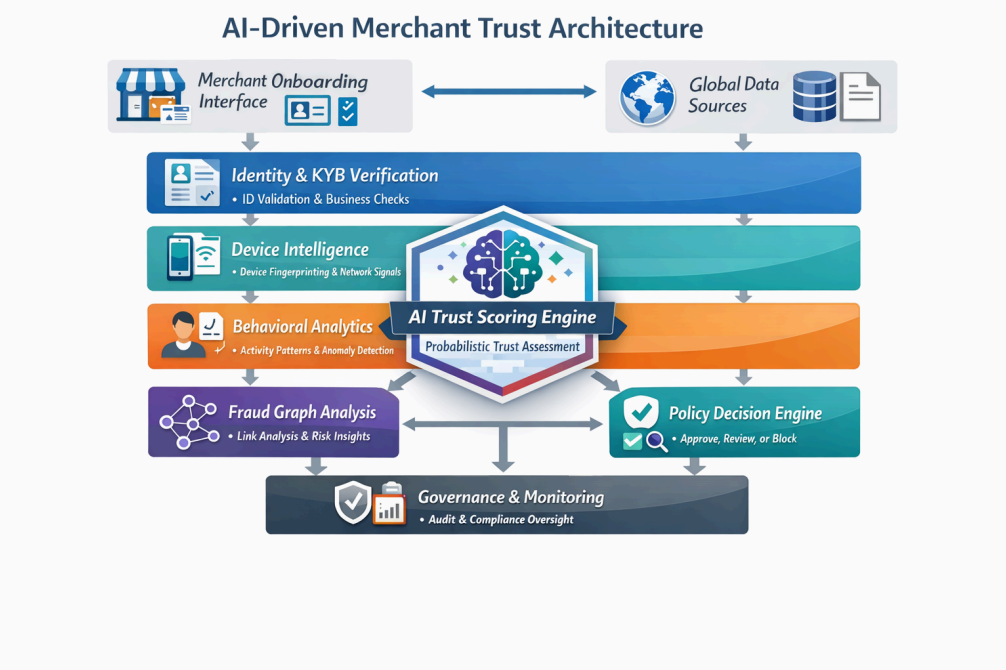


Fig. 1. AI-driven merchant trust architecture: identity and KYB verification, device intelligence, behavioral analytics, and fraud graph analysis feeding a probabilistic trust scoring engine.

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